

Week-1 (Practice Problems)

(1)

User will enter length and width for a rectangle or square, your task is to find the area and the perimeter for the shape.

Bonus: Try to find if the shape is a rectangle or a square.

Input	Output
4 5	Area: 20 Perimeter: 18
7 7	Area: 49 Perimeter: 28

(2)

A student is very bad at converting minutes to seconds, but guess what that student knows a pro coder and that is you. So, now you have to write a program for him so that he can complete his homework. Your program should take minutes (integer) as input from the user and your program should convert that minutes to seconds.

Input	Output
2	2 minutes equals to 120 seconds
30	30 minutes equals to 1800 seconds

(3)

User will enter an integer number and without converting the number to string you need to find the last digit of the user input and the remaining digits.

Input	Output
1048	Last Digit: 8 Remaining Number: 104
111	Last Digit: 1 Remaining Number: 11

(4)

After winning the Fifa World Cup 2022 Messi decided to throw a pizza party. Messi wanted to do something special at this party. So, he decided to create a Pizza slice machine which will take the angle and radius of the pizza slice, after that the machine will tell you the area/size of that specific slice. Now you need to write a code that helps Messi to build this machine.

$$[\text{slice area} = (\text{angle} / 360) * \pi * \text{radius} * \text{radius}]$$

Input	Output
60 8	33.51
45 9	31.81

(5)

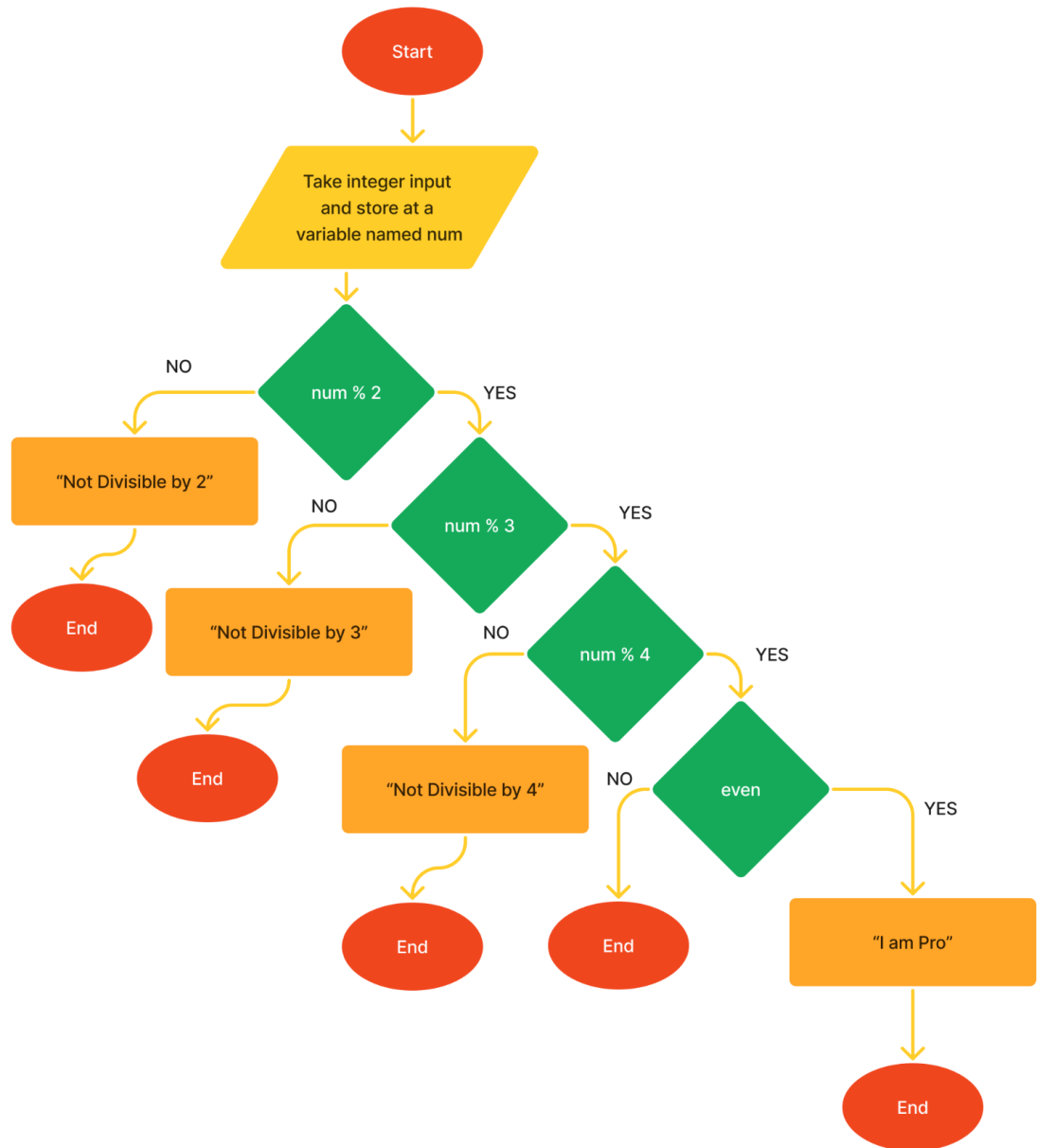
You are given a task to write a code that will greet every cricket player with their name, total runs scored and number of balls faced and number of wickets taken. Furthermore you need to calculate the strike rate of the batsman from the given information.

Input	Output
Tamim Iqbal 2048 1576 0	Congratulations Tamim Iqbal, Here is your stats: Runs Scored: 2048 Balls Faced: 1576 Strike Rate: 129.95 Total Wickets: 0
Mitchell Starc 78 100 125	Congratulations Mitchell Starc, Here is your stats: Runs Scored: 78 Balls Faced: 100 Strike Rate: 78 Total Wickets: 125

Week-2 (Practice Problems) Branching

(1)

Convert the given **flowchart** to python code.



(2)

Users will always enter a **4 digit number** and you need to add the first digit and the last digit of that number and see if the summation is even or not. If the summation is even then print **“Even”** otherwise print **“Odd”**.

[YOU ARE NOT ALLOWED TO USE STRINGS]

Input	Output
1756 Explanation: First digit is 1 and Last digit is 6. Now if we add 1+6 we get 7, and 7 is Odd	Odd
2424 Explanation: First digit is 2 and Last digit is 4. Now if we add 2+4 we get 6, and 6 is Even	Even

(3)

You want to join a Cycling community, to join there you need to enter the distance traveled by cycle in **km**, using that distance you will either get a title or you will be rejected to join the club. The table below shows the title for each distance in **m**.

[HINT: Read the question carefully]

Distance (m)	Result
>10000	“Veteran Cyclist”
>=9000	“Hardcore Cyclist”
>=8000	“Pro Cyclist”
>=7000	“Amateur Cyclist”
>=6000	“Beginner Cyclist”
>=5000	“Rookie Cyclist”
<5000	“You are not Eligible”

Input	Output
12	Veteran Cyclist
4.5	You are not Eligible

(4)

You already have two angles **a** and **b** with values **85** and **35** respectively in two different variables. Now, the user will enter the **third angle**, after that you have to say whether you can form a triangle with these three angles.

Input	Output
60	Triangle can be formed
30	Triangle can not be formed

(5)

Write a program that asks the user for an hour as input and prints a message, which lab class is taking place in the lab based on the following table. If the user enters any value that is not in the table, your program should print **“Wrong Input”**.

Hour (input)	Message (output)
8	1st lab
9	1st lab
10	1st lab
11	2nd lab
12	2nd lab
13	2nd lab
14	3rd lab
15	3rd lab
16	3rd lab

Input	Output
13	2nd lab
16	3rd lab
7	Wrong input

(6)

Suppose the following expressions are used to calculate the values of **P** for different values of **Q**:

$$P = 6000 + 15Q^2 \quad \text{if } Q < 50$$

$$P = \frac{1000}{3+(Q^2/1900)} \quad \text{if } Q \geq 50$$

Write a python code that takes the value of **Q** as an user input and shows the value of **P** as an output.

(7)

Write a python program that asks the user for hours in **integers** and prints the **CEFR Level** according to the table below. If the user enters a number of hours not included in the table below, print “**Unknown**”.

CEFR Level	Number of Hours (Inclusive)
C2	1,000—1,200
C1	700—800
B2	500—600
B1	350—400
A2	180—200

Input	Output
1150	C2
650	Unknown

Loops

(1)

User will enter an integer number and you have to find whether the number is palindrome or not.

[You are not allowed to convert the user input to string]

Input	Output
4728 Explanation: if we reverse the number then it is 8274. Thus the reverse number 8274 is not equal to user input 4729 therefore it is not Palindrome	4728 is not Palindrome
1221 Explanation: if we reverse the number then it is 1221. Thus the reverse number 1221 is equal to user input 1221 therefore it is Palindrome	1221 is Palindrome
4665	4665 is not Palindrome

(2)

You need to help a student for his homework, your task is to write a code that takes two numbers from the user and tell which numbers are prime and which numbers are not prime.

Input	Output
2 5	2 is Prime 3 is Prime 4 is not Prime 5 is Prime
11 23	11 is Prime 12 is not Prime 13 is Prime 14 is not Prime 15 is not Prime 16 is not Prime 17 is Prime 18 is not Prime 19 is Prime 20 is not Prime 21 is not Prime 22 is not Prime 23 is Prime

(3)

It is getting hard to determine who is going to win the Ballon d'Or, so you are hired to write a program to solve the issue. First, an integer number will be given as input which will determine how many players are there. Then you will take information for those players (name, national team, goals scored). Then the Ballon d'Or will be awarded to the player who scored the most goals.

Input	Output
2 Messi Argentina 24 Haaland Norway 56	The Ballon d'Or is awarded to Haaland from Norway for scoring 56 goals this season.
3 Christiano Portugal 10 Chiesa Italy 7 Neymar Brazil 21	The Ballon d'Or is awarded to Neymar from Brazil for scoring 21 goals this season.

(4)

You are given a task to convert the mass of different patients from kg to g. At first the admin will enter how many conversions they need. Then, weight will be entered for each patient and it should convert the weight from kg to g.

Input	Output
3 6 90 55	6000 90000 55000
2 4 22	4000 22000

(5)

There is a series which is known as the Fibonacci series. This series is basically the summation of the previous two preceding numbers. You should write a program that takes an input from the user and print that many numbers of the series.

[Fibonacci Series starts from 0,1]

Input	Output
4	0,1,1,2
8	0,1,1,2,3,5,8,13

(6)

You need to write a code in java that can find the factorial from the user input.

Input	Output
4	24
8	40320

(7)

Write java code for the following patterns:

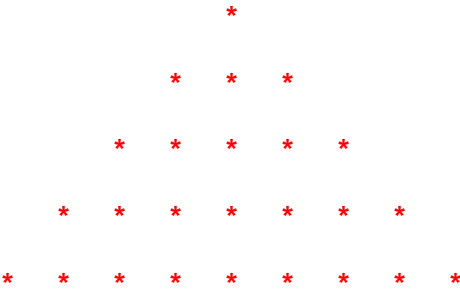
a)

```
* * * * *
*           *
*           *
*           *
*           *
* * * * *
```

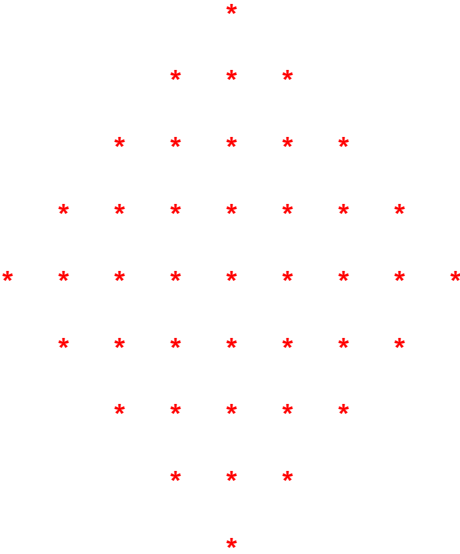
b)



c)



d)



e)

\$	#	\$	#	\$
\$		\$		\$
\$	#	\$	#	\$
\$		\$		\$
\$	#	\$	#	\$

Week-4 (Practice Problems)
Strings

(1)

Write a java code that will take user input as a string and the output should tell whether the input is palindrome or not.

Input	Output
hello Explanation: If we reverse the word hello then we get olleh, now hello is not equal to olleh so it is not palindrome.	hello is not palindrome
madam Explanation: If we reverse the word madam then we get madam, now madam is equal to madam so it is palindrome.	madam is palindrome

(2)

Write a java code that will take one input from the user with numbers separated by comma, your task is add all the prime numbers from that string and display the summation.

Input	Output
2,3,11,6,100,4	16
4,9,122,1000,42	0

(3)

Zark Muckerberg is worried about his security for the messaging system. He wants to ensure that all messages are encrypted before it is sent. Now, Zark Muckerberg hired you to write a code that will encrypt the message. To encrypt first you need to count how many times each letter exists and then change the letter to that many times. For example. "Hello" is the message here H, e, l and o exists 1,1,2 and 1 times respectively. Now, change H to l as H exist only 1 time, e is changed to f as e exists only 1 time, then l is changed to n as l exists 2 times. Thus the final message is "lfnnp".

Input	Output
Hello	lfnnp
zzzxy	ccczzz

(4)

Write a python code that will tell you how many vowels and consonants are there in a single word given by the user.

Input	Output
hello	Vowels: 2 Consonants: 3
ROnaIDo	Vowels: 3 Consonants: 4

(5)

After completing Task:3 Zark Muckerberg was very happy with your work, now he wants to make sure that users will be able to know whether their password is strong or not. He figured out some requirements for a strong password which are: **length ≥ 6** , contains **uppercase**, **lowercase**, **symbols** and **numbers**. Your task is to write a code that will make sure all these requirements are followed, if something is missing then it should what are missing.

Input	Output
password	Uppercase missing Symbol missing Number missing
pass12	Length should be greater than 6 Uppercase missing Symbol missing
P@ssWord123	Strong Password

(6)

Write a code that will take an input from the user and convert all uppercase characters to lowercase and all lowercase characters to uppercase, finally showing the result.

Input	Output
RonALdO	rONaIdO
FreDERiCo CHiesA	fREderlC O chIESa

(7)

First user will enter the name of a football player and then enter the name of a cricket player, then you have to merge the first name of the football player with the last name of the cricket player and show the result.

Input	Output
Christiano Ronaldo Shakib Al Hasan	Christiano Hasan
Frederico Chiesa Mitchell Starc	Frederico Starc

(8)

You are hired to write a program for an elementary school that will make corrections to sentences written by students. Requirements are as follows: first letter will always be capital letter, i will be always uppercase if single and after full stop, exclamation and question mark the next letter should be capital.

Input	Output
i am a football player. the name of my club is Real Madrid.	I am a football player. The name of my club is Real Madrid.
what is your name? i need to know some information about you before you are allowed to register. are you ok with it?	What is your name? I need to know some information about you before you are allowed to register. Are you ok with it?
Messi is the goat. he plays for Argentina. his jersey number is 10. currently he plays for Inter Miami.	Messi is the goat. He plays for Argentina. His jersey number is 10. Currently he plays for Inter Miami.

(9)

You need to write a code in python to develop a calculator that performs addition and subtraction. In the calculator the user will enter a string with operator (+ or -) and you need to display the final result.

Input	Output
5+3+10+2-10	Result: 10
7+100-10+7+3-1	Result: 106
1000+2000-8000-100+5	Result: 5095

Week-5 (Practice Problems)
List

(1)

User will enter a list as a string, your task is to convert the string to a list, if the user enters any number make sure to convert the number to integer.

Note: You are not allowed to use split() function

Input	Output
[7,13,"Hello","Neymar",9.25,"Ronaldo"]	[7,13,"Hello","Neymar",9,"Ronaldo"]
["Hello","Kane","Mabappe",10.88]	["Hello","Kane","Mabappe",10]

(2)

Users will enter a list and a target, your task is to find if there are any two values from the list that can be added together and the target value is achieved. If there is any such case return True otherwise False.

Note: You are not allowed to use split() function

Bonus: Try to do it with 1 loop only

Input	Output
[2,7,3,4,8,11] 9 Explanation: if we add 2 and 7 we can get 9 therefore the output is True	True
[13,4,9,0,8] 2 Explanation: There is no such combination that can be added and 2 can be achieved therefore the output is False	False

(3)

There are two lists, where one contains the name of the students and the other contains marks of each student. Your task is to find the name of the student who got the highest and second highest.

Input	Output
name = ["Ronaldo", "Neymar", "Messi", "Mbappe"] marks = [17,19,9,16]	Highest: Neymar Second Highest: Ronaldo
name = ["Shakib", "Tamim", "Starc", "Morgan"] marks = [2,14,13,9]	Highest: Tamim Second Highest: Starc

(4)

Users will enter a list and your task is to say whether the list is palindrome or not.

Note: You are only allowed to use only one list and you can use split() function

Input	Output
[1,2,3,4,3,2,1]	The list is palindrome
[7,8,8,7]	The list is palindrome
[4,4,4,5,6,4,4,4]	The list is not palindrome

(5)

You are given a list of distance in km, your task is to create a list of distance in m. However the output list should be in reverse order.

Input	Output
[4,13,19,7,12]	[12000,7000,19000,13000,4000]
[0.6,0.085,400,20]	[20000,400000,85.0,600]

(6)

As the election is near, you are given a task to write a code for a digital voting system. You are given a list of markers and their vote count (initially 0). Then there will be another input which will determine how many people can vote. People will choose the marker and based on the input vote count will increase. Finally, show the winner of the election and show the number of votes achieved by each markers.

Input	Output
markers = [{"X":0}, {"Y":0}, {"Z":0}] 5 X X Z Z Z Explanation: X got 2 votes, Y got 0 votes and Z got 3 votes, therefore the winner is Z	Z won the election X got 2 votes Y got 0 votes Z got 3 votes
markers = [{"X":0}, {"Y":0}, {"Z":0}, {"Q":0}] 8 Q Q Z Q X Y X Z	Q won the election X got 2 votes Y got 1 vote Z got 2 vote Q got 3 votes

(7)

Write a python program that will take numbers as input until the user enters "Done". Finally show all the numbers that are prime numbers in a list.

Input	Output
7 4 11 18 Done	[7,11]
19 25 24 13 29 37 40 Done	[19,13,37]

(8)

Messi is worried about his son as he cannot calculate distance between two coordinates, so he requested you to create a calculator for him so that he can pass the exam with ease.

Input	Output
[[4,9],[-2,13]]	7.21
[[11,19],[-7,-19]]	42.04

(9)

Users will always enter a list that contains exactly 5 elements, your task is to find if the sum of the first 4 elements and product of last to element is equal. split() can be used.

Input	Output
[1,3,2,2,4]	Equal Sum: 8 Product: 8
[7,3,2,4,3]	Not Equal Sum: 16

	Product: 12
--	-------------

(10)

After helping Messi's son Ronaldo requested you to help his son by writing a program that will help his son to find mean, median and mode from the numbers entered. You can assume numbers will always be in sorted order.

Input	Output
2,2,2,4,3,5,5,7	Numbers List: [2,2,2,4,3,5,5,7] Mean: 3.75 Mode: 2 Median: 3.5
2,2,2,4,3,5,5	Numbers List: [2,2,2,4,3,5,5] Mean: 3.28 Mode: 2 Median: 4

Task 1

Write a python program that prints all the Fibonacci numbers from 0 to N where N will be given as input by the user.

A Fibonacci number is a number which is the summation of its previous two Fibonacci numbers.

The first two Fibonacci numbers are 0 and 1.

So, the 3rd is $0 + 1 = 1$,

the 4th is $1 + 1 = 2$,

the 5th is $1 + 2 = 3$,

the 6th one is $2 + 3 = 5$,

and so on.

i.e. 0, 1, 1, 2, 3, 5, 8, 13, 21.....

=====

==Sample Input 1:

10

Sample Output 1:

0 1 1 2 3 5 8

Explanation: All Fibonacci numbers up to the number 10 is printed separated by spaces.

=====

==Sample Input 2:

15

Sample Output 2:

0 1 1 2 3 5 8 13

Explanation: All Fibonacci numbers up to the number 15 is printed separated by spaces.

Task 2

Write a Python program that asks the user for a range (a starting number and an ending number) and then **counts** how many numbers are prime numbers and how many numbers are perfect numbers between that range. It also **prints** those numbers in the format shown in the output of the example given below.

=====

Hint (1): We may declare two strings/lists to store the prime and perfect numbers. Inside the loop, we check for prime and perfect numbers and add them to respective string/list besides counting them.

Hint (2): We may need to convert numbers in String to integers for checking and calculations. We may again need to convert numbers to Strings for storing and printing as the outputs shown in the example below.

=====

Example: If the user gives us a range of between 2 and 6, there are 3 prime numbers (2, 3, 5) and 1 perfect number (6) in that range.

Input:

2

6

Output:

Between 2 and 6,

Found 3 prime numbers

Found 1 perfect number

Prime numbers: 2, 3, 5

Perfect number: 6

Task-3

Imagine you are playing a game with your friend which is quite similar to ludo. The player who crosses

throughout the board of 25 boxes wins. Each of you gets to roll the dice and can get from 1-6.

If both players converge on the same box, the player's avatar coming to this box previously will get eaten and

the player will lose.

There are certain checkboxes in the board where the avatars cannot be eaten. The checkpoints are 10th and

20th.

Until a player's avatar gets eaten or he/she crosses the 25th box, the dice will be rolled between the players

and the game will continue.

Sample Input 1: Player 1: 6 Player 2: 4 Player 1: 3 Player 2: 5 Sample Output: Player 1 died Player 2 wins	Sample Input 2: Player 1: 2 Player 2: 3 Player 1: 1 Sample Output: Player 2 died Player 1 wins
Sample Input 3: Player 1: 5 Player 2: 4 Player 1: 5 Player 2: 6 Player 1: 6 Player 2: 3 Player 1: 4 Player 2: 3 Player 1: 5 Sample Output: Player 1 wins	

TASK-4

Rhombus, Just draw the image of the above triangle once. And then, the opposite, once.

Sample Input: 4

Sample Output :

			1			
		1		3		
	1				5	
1						7
	1				5	
		1		3		
			1			

TASK-2

Write a python program that prints alphabet pattern 'Z' of size N using * where N will be given as input.

Sample Input1: 5

Sample Output1:

	C1	C2	C3	C4	C5
L1	*	*	*	*	*
L2				*	
L3			*		
L4		*			
L5	*	*	*	*	*

Sample Input2: 4

Sample Output2:

	C1	C2	C3	C4
L1	*	*	*	*
L2			*	
L3		*		
L4	*	*	*	*



Task-1 : Write a python program using a multiple variable and multiple value concept which reads 1 integer value, 1 float value and 1 string value and prints them

Sample input:

1 1.0 'T'

Sample output:

11.0T

Task-2

Write a python program using multiple variable and one value concept which reads a string number

Sample input:

1

Sample Output:

1

Sample input2:

The

Sample output2:

The

Question 1: What is the output of the following code?

```
>>> x=100
>>> y=100
>>> id(x) == id(y)
```

- | | |
|------|-------|
| I. | True |
| II. | False |
| III. | Error |

Question 2: Which of the following would give a syntax error?

- | | |
|------|---|
| I. | <pre>>>> x, y, z = 10, 20, 30</pre> |
| II. | <pre>>>> x, y, z = 10, 'Hello', True</pre> |
| III. | <pre>>>> x = 10, y = 'Hello', z = True</pre> |
| IV. | None of the above |

Question 3: Which of the following statements are correct?

1) The variable names cannot start with a digit.
2) Variables names in Python are case-sensitive.
3) Variable names cannot start with the underscore _.
4) Variable names can be reserved keywords.

Question 5: Which of the following is true?

I.	$9\%2=4$
II.	$9//2=4$
III.	$9/2=4$
IV.	None of the above

Project

Before your start kindly visit the link below it will guide you with some python packages that are very simple but very effective and can help you with below projects and also help you out in long run. There are in total 3 ideas arranged in increasing difficulty.

[Google Colab Link](#)

Note: For all the ideas below try to implement them with using learnings from functions

Idea 1 (EASY)

Try to create a guessing game, where a dice will be rolled 5 times and the player needs to guess the number. If the player can guess the number correctly then the player gets 1 point. Player will win when 3 point is reached otherwise the ai or computer will win.

Idea 2 (Medium)

Create an expense tracker that will keep record of money that you earned and spend. For expense there will be four categories food, travel, friends and others. For income there will be 3 categories pocket money, job and others. This looks simple right? Don't worry there is more things to come. You should create a function that can generate a report whenever it is called, this report will show what expenses and income you made in each category and finally total expense and result.

Let me help you out with what functions do you need for this:

1. **add_expense(category, amount)**
2. **add_income(category, amount)**
3. **generate_report()**

You are most welcome to add more features to it but the above 3 features are must

Idea 3 (Hard)

I am pretty sure everybody watched Squid Game and the marble game was the best game in my opinion, therefore let's try to recreate that game with python. And btw if you did not watch the series I'll explain the game below so that you can get your hand dirty with python code.

There will be two players (One you and another AI), both will have 10 marbles to start with. This game will be in turns. There will be a coin toss which will always be called by the player. If the player wins then he or she can choose whether to guess first or not. If the player chooses to guess first then the AI will make a bet of any amount of marbles that it has. After the bet is done the player will guess how much marbles did the AI bet. If the player is correct then AI will lose that amount of marble that he betted, on otherhand if the player makes a wrong guess then the player needs to give the amount of marble that is betted by the AI. In next turn the player will bet and the AI will guess. In this way the game will continue until one of the player don't have any marbles.

Constraints:

Each player can bet only the amount of marble he or she has. For example if a player has only 3 marbles then he or she can only bet maximum 3 marbles not more than that, but he or she is welcome to bet less than 3 but greater than 0.

If you want to make this game more interesting then you are welcome to add twists to it by adding new rules.

FINAL MOCK

Answer all the questions below

(Note: You will get two coding problems and one tracing problem in your real final exam.)

(1)

Write a Python function that takes an integer number as input from the user. Without converting the number to a string, find the reverse of the string of the number, but keep only the even digits, and also find the sum of all the even numbers.

Input	Output
4356	Reverse: 64 Sum: 10
12345984	Reverse: 4842 Sum: 18

(2)

Assume you are given two dictionaries. The dictionary named `current_sem_data` contains GPA of 3 students of their current semester courses. Another dictionary named `prev_sem_data` contains their previous semesters CGPA and credits completed respectively. Assume all courses are of 3 credit points each.

```
current_sem_data = {  
    "Messi": (4.0, 3.7, 4.0),  
    "Ronaldo": (3.3, 3.7, 4.0),  
    "Neymar": (3.3, 3.7, 4.0)  
}
```

```
prev_sem_data = {  
    "Messi": (3.88, 90),  
    "Ronaldo": (3.52, 70),  
    "Neymar": (3.25, 85)  
}
```

Now write a python programme that will take a student's name as user input and calculate and print that student's current CGPA using both dictionaries.

Formulas:

$\text{current_total_GPA} = \sum \text{current_course_grade} * \text{credit}$

$\text{previous_total_GPA} = \text{previous_CGPA} * \text{credits_completed}$

$\text{CGPA} = (\text{current_total_GPA} + \text{previous_total_GPA}) / \text{Total_credits_completed}$

Input	Output
Neymar	####Data for Neymar#### Prev Sem Credit: 85 Prev Sem CGPA: 3.25 Credits Completed with Current Sem: 94 Current Sem GPA: 33.0 CGPA: 3.29

(3)

```
players = {
    "Benzema": {
        "Ovr": 99,
        "Pos": "CF",
        "Pac": 97,
        "Sho": 99,
        "Pas": 93,
    },
    "Lewandoski": {
        "Ovr": 97,
        "Pos": "ST",
        "Pac": 90,
        "Sho": 99,
        "Pas": 86,
    },
    "Tonali": {
        "Ovr": 92,
        "Pos": "CDM",
        "Pac": 90,
        "Sho": 85,
        "Pas": 91,
    }
}
```

You are given a player's dictionary named `players` that contains some player information from FIFA Ultimate Team (Fifa 23). Your task is to write a Python function that must take the player name as a parameter and have different attributes. Then you have to create a new dictionary with the information about that player and his attributes and finally return the new dictionary. If no attribute is entered, return a dictionary with the player name and all the attributes.

Input	Return
func("Tonali")	{'Tonali': {'Ovr': 92, 'Pos': 'CDM', 'Pac': 90, 'Sho': 85, 'Pas': 91}}
func("Benzema", "Pos", "Sho")	{'Benzema': {'Pos': 'CF', 'Sho': 99}}
func("Lewandoski", "Pac", "Pos", "Pas")	{'Lewandoski': {'Pac': 90, 'Pos': 'ST', 'Pas': 86}}